Twenty-Metre WSPR Propagation Observed from a Mobile 200 mW Beacon in Southern Finland

Diurnal, geographic, seasonal and space-weather patterns over three summer seasons, 2024-07-04 to 2026-11-06

Otso Laakso, OH2GAX

WSPR Logger project note — version 1.1 — 6 November 2026 — `github.com/oh2gax/wspr_logger`

Abstract

A 200 mW WSPR beacon on 14 MHz (callsign OH2GAX, Diamond monoband whip on a car roof, home locator KP20jg) has been logged by the WSPR Logger software. This note analyses three summer seasons (2024, 2025 and 2026): 495 calendar days with data between 2024-07-04 and 2026-11-06, 54 868 poll snapshots and 3 092 119 individual reception reports from 3 713 distinct receiving stations. The first two seasons were rebuilt from the public `wspr.live` database; the solar and ionospheric indices (SFI, K and A index, X-ray class, Juliusruh MUF and foF2) exist only for the 166 days logged live in 2026 (2026-05-08 to 2026-11-06), and the space-weather analysis is confined to them. The band is effectively closed between about 00 and 04 UTC (median 4 stations or fewer) and carries a median of about 75 stations between 08 and 16 UTC. 83.5 % of the receiving stations are in Europe and 75 % lie between south and west-south-west of the transmitter. At least one station farther than 6 000 km was heard on 360 of 495 days (73 %), and one farther than 10 000 km on 167 days; the record is 15 619 km (VK7JJ, 2024-07-10), i.e. 78 095 km per watt. Long-distance windows are concentrated to the early morning and late morning (North America, 04–07 UTC, with a second maximum near 10 UTC) and late afternoon (Africa and Oceania, 16–19 UTC). Daytime reporting is lowest in midsummer and highest in autumn: in 2026 the median fell from 85 stations in May to 32 in July and rose to 110 in October. In the months observed in all three seasons (August, September, October) the median daytime count was 60, 89and 96 stations in 2024, 2025 and 2026, and the share of snapshots with a station beyond 6 000 km was 17.9, 19.3 and 13.6 %. In the live-logged 2026 data, after removing the diurnal rhythm, reporting is strongly tied to the Juliusruh MUF (index 27 for MUF $<$ 12 MHz versus 144 at 20–24 MHz) and weakly and negatively to the geomagnetic K-index (correlation -0.16); the 10.7 cm solar flux alone explains nothing ($r = 0.05$). The worst day of the live-logged data (4 July 2026, K = 7, A = 74, 10 stations against a typical 80) followed an M-class flare period with SFI above 195.

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1 Introduction

The Weak Signal Propagation Reporter (WSPR) protocol turns every amateur transmitter into a propagation probe: a two-minute, 6 Hz-wide signal is decoded by volunteer receivers around the world and each decode is stored with the receiver position, signal-to-noise ratio (SNR) and distance [1, 3]. A 200 mW beacon is a modest probe, but because the same transmitter transmits on the same schedule every day, the stream of receptions is a long, homogeneous time series of the 20 m band as seen from one place.

The WSPR Logger started as a live dashboard for such a beacon: it polls the public WSPR database `wspr.live` [2] every ten minutes, draws the reporters on a map, keeps a log of the farthest reporter per transmission and records the solar conditions at the time. A second database, the *replay database*, stores every poll snapshot in full. The present note is the first systematic analysis of that archive, extended by a rebuild of the two earlier seasons from `wspr.live`. It asks four questions:

(a) When is the band open, and when are long-distance paths available?

(b) In which directions and at which distances does a mobile 200 mW beacon in Finland actually get heard?

(c) How does reporting change over the season?

(d) How much of the variation is explained by the space-weather indices the logger records?

The note is deliberately descriptive. Where a number can be given with a confidence interval it is; where the design of the data collection limits what can be concluded, this is stated (Section 5).

2 Station, data and definitions

2.1 Transmitting station

The transmitter is a ZachTek WSPR desktop transmitter [4] with built-in GPS (used for timing and for the position), producing about 200 mW (23 dBm) into 50 Ω. The antenna is a Diamond monoband 14 MHz whip on a magnetic mount on the roof of a VW Touran. The home locator is KP20jg (60.27°N, 24.79°E), but the car moves; 86 % of the 42 723 logged transmissions carry the home locator and 102 different six-character locators occur in total (the mobile aspect is the subject of a companion paper on position reporting). One WSPR transmission is made every ten minutes.

2.2 Reception data

Every ten minutes (at :08, :18, . . .) the logger queries `wspr.live` for the receptions of the last 45 minutes and stores, for the current *snapshot*, every station that has reported the beacon within the preceding 60

minutes: callsign, Maidenhead grid, SNR (dB, 2 500 Hz reference bandwidth) and great-circle distance to the transmitter. The live replay archive starts on 2026-05-08. The two earlier seasons (from 1 July 2024 to 7 May 2026) were *rebuilt*: the script `tools/backfill.py` downloads every reception of the callsign on 20 m from `wspr.live` and reconstructs, for each poll time, the list of stations heard in the preceding 60 minutes. Winter months in which the transmitter was not used simply contain no snapshots. The result is 54 868 snapshots on 495 calendar days (82, 188 and 225 days in 2024, 2025 and 2026); within the days with data the snapshots cover about 77 % of the 24 hours, the rest being times with no report or with the logger offline. The replay database holds 3 092 119 rows. A second, older database stores one record per heard transmission (42 723 records on 490 days) with the number of reporters and the largest distance.

The rebuilt snapshots are not guaranteed to equal what the live poll would have stored. The live poll sees only the reports that had reached `wspr.live` when it ran, and distant stations tend to upload later (Section 5); the rebuild, made afterwards, also sees those late reports, so rebuilt snapshots could in principle list more stations, especially distant ones. This was measured: the live-logged days 1 June 2026 to 7 June 2026 were rebuilt from `wspr.live` and compared with the stored snapshots. 995 snapshots matched (one stored snapshot had no rebuilt counterpart, none the other way); the mean number of stations per snapshot was 41.0 stored against 41.0 rebuilt (ratio 1.001), the rebuild listed more stations in 3 % of the snapshots and never fewer, and a station beyond 6 000 km was present in 9.8 % of the snapshots in both. The effect of late reports is therefore negligible for the statistics below, and differences between seasons are not an artefact of the rebuild (the check covers one week of one season, so it is evidence, not proof, for the others).

2.3 Solar-terrestrial data

At every live poll the logger stores the 10.7 cm solar flux (SFI), the planetary K and A indices, the GOES X-ray class, and the maximum usable frequency (MUF, 3 000 km reference) and critical frequency foF2 of the Juliusruh ionosonde (54.6°N, 13.4°E) [5, 6]. Indices come from NOAA SWPC-derived feeds [7, 8]. These readings exist for the 18 885 snapshots of the 166 live-logged days (2026-05-08 to 2026-11-06); the rebuilt seasons carry none, so everything that involves space weather (Section 4.7 and the case study) is limited to 2026. Readings of zero in SFI or MUF are feed failures (76 snapshots) and are treated as missing. foF2 is available for only 16 % of those snapshots (2 973) because the earlier foF2 source was unreliable; it is not used quantitatively here.

2.4 Definitions

Reporters n_t
number of distinct stations that reported the beacon in the 60 minutes before snapshot t.

Expected reporters $\bar{n}_h$
mean of n_t over all snapshots in UTC hour h (the typical value for that hour); in the space-weather analysis the mean is taken over the live-logged snapshots only.

Index
$100 \sum n_t / \sum \bar{n}_{h(t)}$ over a set of snapshots. A value of 100 means "typical for that hour of the day"; this removes the strong day–night rhythm, which would otherwise masquerade as a solar effect.

DX window
snapshot in which at least one reporter was farther than 6 000 km. DX share is the percentage of snapshots that are DX windows.

Daytime
08–16 UTC (10–18 or 11–19 local time, depending on daylight saving).

2.5 Quality control

Each (callsign, grid) pair was compared with the distance the logger stored: a pair is rejected if the distance computed from the receiver's grid differs from the reported distance by more than 500 km. Of 5041 pairs, 3 were rejected: `G8LZI/2` (grid JO02gl, listed at 11 643 km, grid 1 687 kmaway); `SM4HVP` (grid JP70od, listed at 17 038 km, grid 538 kmaway); `SV1CDN` (grid KM17tx, listed at 5 034 km, grid 2 535 kmaway). Their 5 737

rows were dropped from every statistic below. Without this step the farthest "reporter" of the whole record would have been spurious. All station and distance results use the cleaned data.

3 Methods

Confidence intervals. Proportions of snapshots (DX shares) carry 95 % Wilson intervals. Indices and correlations carry 95 % day-block bootstrap intervals (2 000 resamples of whole days with replacement), because consecutive snapshots share 50 of their 60 minutes and are strongly autocorrelated; treating them as independent would make every interval far too narrow.

Regions. Receiving stations were assigned to a region from the centre of their four-character grid. **Sectors.** Bearing from the transmitter home locator is divided into eight 45° sectors. **Distance bands.** 0–1 500, 1 500–3 000, 3 000–6 000, 6 000–10 000 and >10 000 km. A *station* statistic uses the median over the snapshots in which that station appears.

Space-weather bins. MUF, K and A are binned and, within each bin, the index and DX share are computed over all snapshots in the bin. Bins with fewer than 30 snapshots are omitted. Correlations use the per-snapshot index $n_t/\bar{n}_{h(t)}$ (relative reporting) and the raw n_t for comparison.

Reproducibility. All figures and numbers in this note are generated by `analyze.py` from the two databases; the LaTeX macros in the text are written by the script, so no value is typed by hand (Appendix A).

4 Results

4.1 Overview

The record contains 54 868 snapshots from 2024-07-04 to 2026-11-06, with a mean of 56.3 and a median of 53 reporters per snapshot; 3 713 distinct callsigns appear (5 038 callsign–grid pairs). 73 % of days (360 of 495) had at least one reporter beyond 6 000 km, 167 days (34 %) beyond 10 000 km. 16.3 % of all snapshots are DX windows. Figure 1 shows the structure at a glance: three seasons separated by the winter break, a regular daytime band of strong reporting, a dead band in the small hours, occasional dropouts of the logger, a weaker midsummer, and a brightening in September–October. The seasons are compared in Section 4.6.

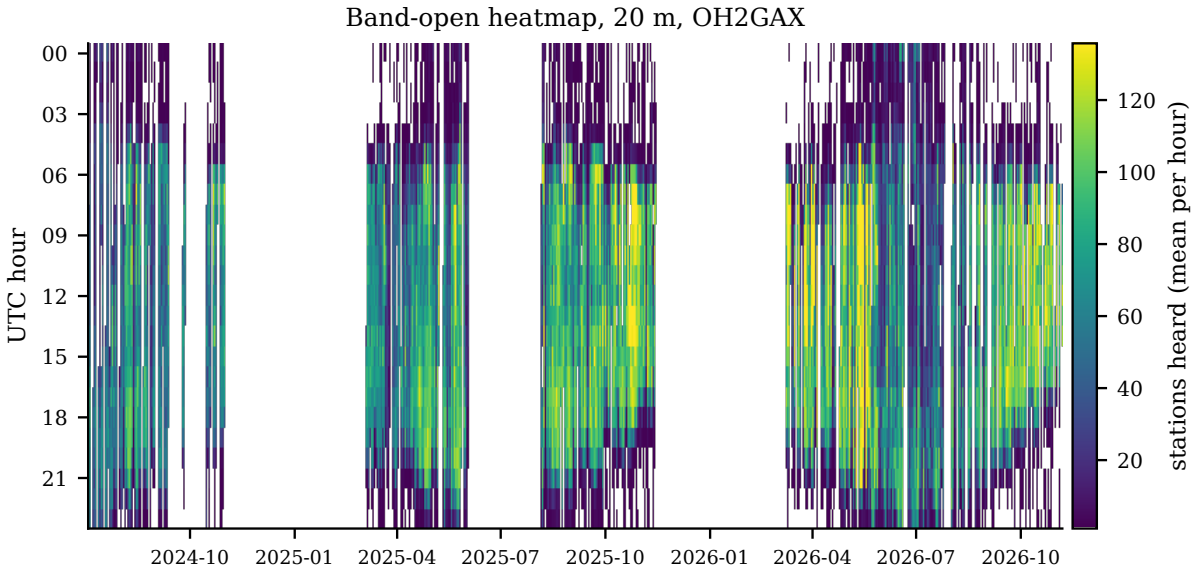


Figure 1. Band-open heatmap for 20 m: mean number of stations heard per UTC hour for each calendar day (colour clipped at the 97th percentile). White columns are days without data: the winter break, days on which the transmitter was off, and periods when the logger was not running.

4.2 Diurnal pattern

Figure 2a shows the typical day. Between 00 and 04 UTC the median is ≤4 stations; reporting rises steeply after 05 and the median is at least 50 stations from 7 to 20 UTC; from 08 to 16 UTC it is about 75 (maximum hourly median 81). The interquartile range is wide, on average 46 to 99 stations between 08 and 16 UTC, so day-to-day variability is large compared with the typical value.

The probability of a long-distance window is *not* highest when the band is busiest (Figure 2b). The share of snapshots with a reporter beyond 6 000 km peaks at 28 % around 5 UTC but is only 7 % at 14 UTC; 28 % at 05 UTC and 11 % at 19 UTC. A reporter beyond 3 000 km is available in up to 58 % of snapshots at 6 UTC and in only 22 % at 13 UTC. A plausible reading is that around noon the F2 layer carries the short and medium paths into Europe while the long paths, which need the layer to be illuminated at both ends or along a grey line, are least likely; the morning and evening maxima coincide with the periods when the path to North America (morning) and to Africa and Oceania (afternoon and evening) passes through twilight. The logger data cannot test this mechanism; they only document the timing.

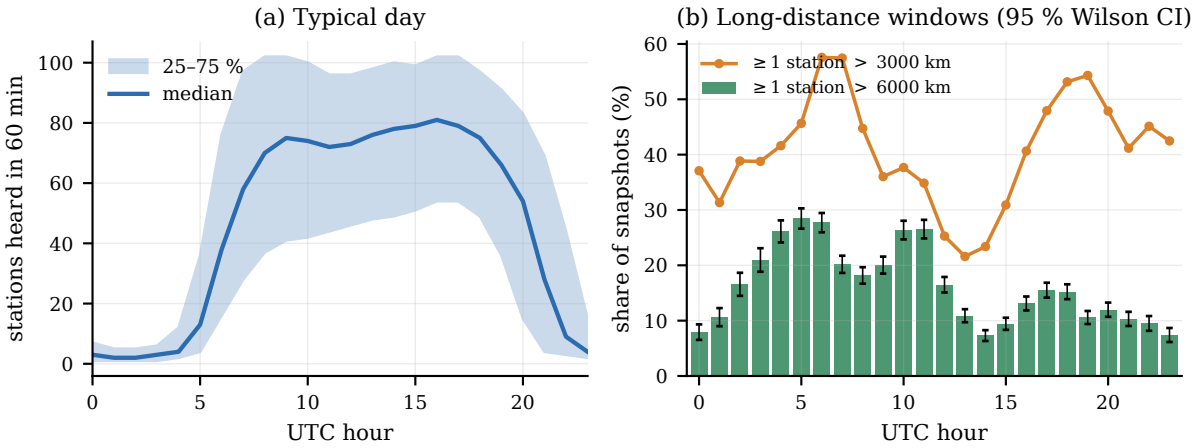


Figure 2. (a) Typical day: median and interquartile range of the stations heard in 60 minutes by UTC hour. (b) Share of snapshots with at least one station beyond 6 000 km (bars, 95 % Wilson intervals) and beyond 3 000 km (line).

4.3 Geography: where the beacon is heard

Table 1 and Figure 3 show that the footprint is overwhelmingly European: 4207 of 5 038 stations (83.5 %). North America contributes 120 stations (2.4 %), Africa 60, Oceania 25 and South America 7; Asia only 9. Measured by bearing, 2865 stations lie in the south-west sector (225–270°), 895 in the south sector (180–225°), 256 to the west and 168 to the south-east; the east and north together hold only about 2 % and 15 % of the stations. The receiver network is densest in Central and Western Europe, which are also the directions of the best 20 m paths from Finland, so this is a joint property of propagation and receiver density, not of propagation alone.

The reporting is also concentrated on a few stations: 88 stations account for half of all reports (the three most frequent are G4ZFQ, ON5KQ, OE9GHV; the top station appears in 39 671 snapshots). These are mostly continuously running receivers, so the network response to a change in propagation is dominated by a small set of fixed receivers.

4.4 Distance and signal strength

Over all reports the median SNR is -19 dB with quartiles at -24 and -14 dB (5th percentile -28 dB); the lowest value in the archive is -35 dB and the highest +24 dB, measured at `DK6UG` at 1 592 km on 2026-09-17 (Table 2). SNR falls with the logarithm of distance, but weakly (correlation -0.19 between $\log_{10} d$ and the station median SNR): the median is -20 dB below 1 500 km, -21 to -22 dB between 1 500 and 6 000 km

Table 1. Receiving stations by region (unique callsign–grid pairs, quality-controlled).

Region of station	Stations	Share (%)	Median dist. (km)	Max. dist. (km)	Best SNR (dB)
Europe	4 207	83.5	1 595	4 682	+24
North America	120	2.4	6 868	9 102	+1
Africa	60	1.2	3 160	9 877	+6
Oceania	25	0.5	15 070	15 619	+16
South America	7	0.1	11 351	13 421	-5
Asia	9	0.2	2 470	4 466	-1
Other / unknown	610	12.1	2 305	14 785	-5

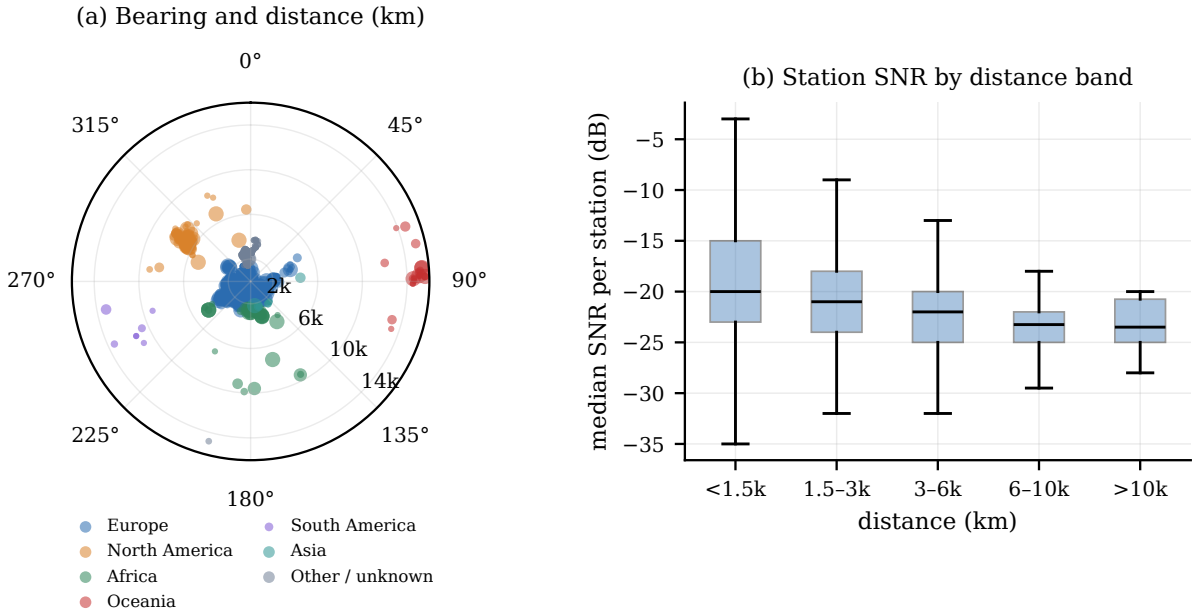


Figure 3. (a) Position of every receiving station by bearing from the transmitter and distance (dot size grows with the number of snapshots in which the station was heard; colour = region). (b) Median SNR per station by distance band (boxes: quartiles; whiskers: 1.5 interquartile ranges).

and only -23 to -24 dB beyond 6 000 km. The spread narrows with distance (interquartile range 8 dB below 1 500 km, 3 to 4 dB beyond 6 000 km), and the long-distance stations lie close to the lower decoding limit of the archive, about −33 dB, so the long paths operate with little margin and a few dB of extra attenuation removes them. The weak distance dependence is what one expects for a single F2 hop whose loss is dominated by absorption and focusing rather than by free-space spreading; the logger data do not test this explanation.

Table 2. Station SNR by distance band (median over the snapshots in which a station was heard).

Distance (km)	Stations	Median SNR (dB)	Q1 (dB)	Q3 (dB)	Best SNR (dB)
< 1 500	1 545	-20	-23	-15	+22
1 500–3 000	3 235	-21	-24	-18	+24
3 000–6 000	102	-22	-25	-20	+1
6 000–10 000	124	-23	-25	-22	+1
> 10 000	32	-24	-25	-21	+16

4.5 Long-distance windows by region

Long-distance reports (>6 000 km) come from three regions. North America reported on 290 days and shows two maxima (Figure 4): one in the early morning, up to 31 % of days at 5 UTC, and a second in the late morning, 33 % of days at 10 UTC, separated by a dip to 15 % at 8 UTC; from 14 UTC onwards it never exceeds 9 %. Africa reported on 76 days, concentrated in the afternoon and evening (16–19 UTC; maximum 8 % of days at 17 UTC). Oceania reported on 161 days, with a late-afternoon maximum (12 % at 17 UTC) and a smaller early-morning one (11 % at 7 UTC). South America contributed on 12 days only. The farthest of each region is in Table 3.

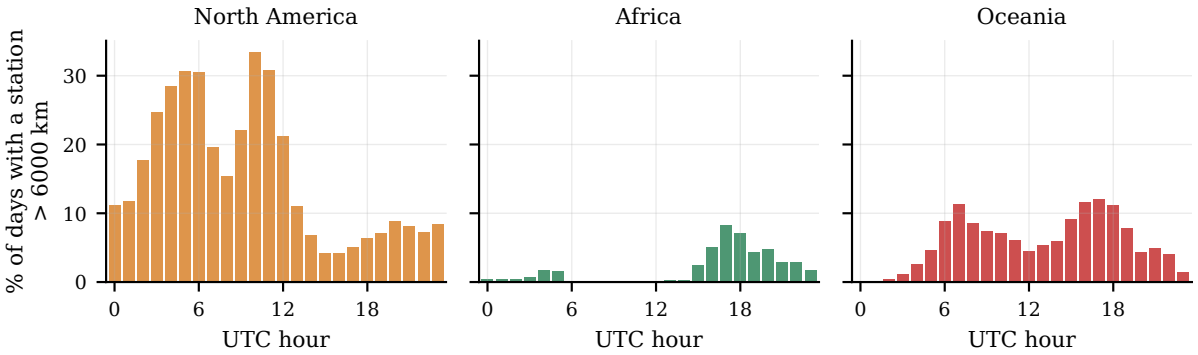


Figure 4. Percentage of days with at least one station of the region beyond 6 000 km in each UTC hour, for the three regions that produced long-distance reports.

Table 3. Days with at least one station beyond 6 000 km, by region.

Region	Days	Best 3-h window (UTC)	Peak share (%)	Farthest (km)	Station, date
North America	290	04–07	33	9 102	YV5RAB, 2026-09-24
Africa	76	16–19	8	9 877	ZS3Y, 2026-06-25
Oceania	161	16–19	12	15 619	VK7JJ, 2026-06-23
South America	12	–	–	13 421	LU3DJ, 2025-09-22

Share = percentage of days with data in that UTC hour on which at least one station of the region beyond 6 000 km was heard.

4.6 Seasonal variation and the three seasons

The archive covers three seasons of unequal length (Table 4): July–October 2024 (82 days with data), March–November 2025 (188 days) and March–November 2026 so far (225 days). The transmitter is not

used in winter, and in 2025 almost nothing was logged between early June and the end of July. The seasons are therefore compared month by month (Table 5, Figure 5).

Within a season. Monthly daytime reporting (Table 6, Figure 6) shows a pronounced pattern that is *not* the one the solar flux alone would predict. In 2026, the only season with a spring-to-autumn record, the median daytime count was 96 in March, 85 in May, fell to 32 in July and rose to 110 in October. The midsummer trough is also seen in 2024 (42 in July against 62 in August); in 2025 the midsummer months were not logged, but reporting rose from 61 in March to 103 in October. The mean SFI of 2026 went the other way: 121 in May, 139 in June and 133 in July, down to 110 in October. The correlation of the monthly median reporting with the monthly mean SFI is -0.88 and with the monthly median Juliusruh MUF 0.92 (6 complete months with solar data, so these are descriptive, not statistical, values).

Two effects probably contribute and cannot be separated in this record. First, the mid-latitude F2 layer has a well-known seasonal anomaly: for the same solar activity the daytime critical frequency is higher in the winter half-year than in summer, which raises the MUF in autumn (here from a daytime median of 19.8 MHz in May and 18.1 MHz in July to 24.4 MHz in October and 26.8 MHz in November 2026). Second, the receiving network itself changes: the number of distinct stations that heard the beacon fell from 745 in May 2026 to 514 in July and rose to 765 in October, which is consistent with fewer attended and automatic receivers during the northern-hemisphere summer. Without an independent reference beacon, neither explanation can be ruled out.

Between seasons. In the months observed in all three seasons (August, September, October) the median daytime count was 60 stations in 2024, 89 in 2025 and 96 in 2026. The step from 2024 to 2025 coincides with a rise in the number of distinct stations heard in the same months (445 in October 2024 against 773 in October 2025, 660 against 755 in August); it is probably a mixture of a growing receiver network, a different state of the ionosphere (solar cycle 25 passed its maximum within the period, but the flux of 2024 and 2025 is not archived), the bias of the rebuilt snapshots (Section 5) and possible changes of the station that the data cannot show. August 2026 is the exception to the upward trend (65 against 82 in 2025). The long-distance share did not follow the daytime count: 17.9 % of the snapshots of these months had a station beyond 6 000 km in 2024, 19.3 % in 2025 and 13.6 % in 2026. Every season produced reports beyond 15 000 km (Table 4).

Table 4. The three seasons. Daytime median = median number of stations heard between 08 and 16 UTC; stations = distinct callsigns heard in the season; DX share = snapshots with a station beyond 6 000 km; days with DX = days with at least one such station. The seasons cover different months (Table 5).

Season	Days	Snapshots	Stations	Daytime median	DX share (%)	Days with DX (%)	Farthest (km)
2024	82	8 730	1 098	55	15.5	71	15 619
2025	188	21 424	2 045	79	14.6	66	15 619
2026	225	24 714	2 112	82	18.1	79	15 619

Table 5. Month-by-month comparison of the seasons: median number of stations heard between 08 and 16 UTC, with the DX share in per cent in parentheses. Months with fewer than ten days of data are left out (–).

Month	2024	2025	2026
Mar	–	61 (5)	96 (31)
Apr	–	65 (13)	90 (25)
May	–	73 (16)	85 (20)
Jun	–	–	49 (16)
Jul	42 (9)	–	32 (16)
Aug	62 (21)	82 (22)	65 (13)
Sep	57 (16)	85 (17)	99 (14)
Oct	61 (13)	103 (19)	110 (14)
Nov	–	92 (1)	–

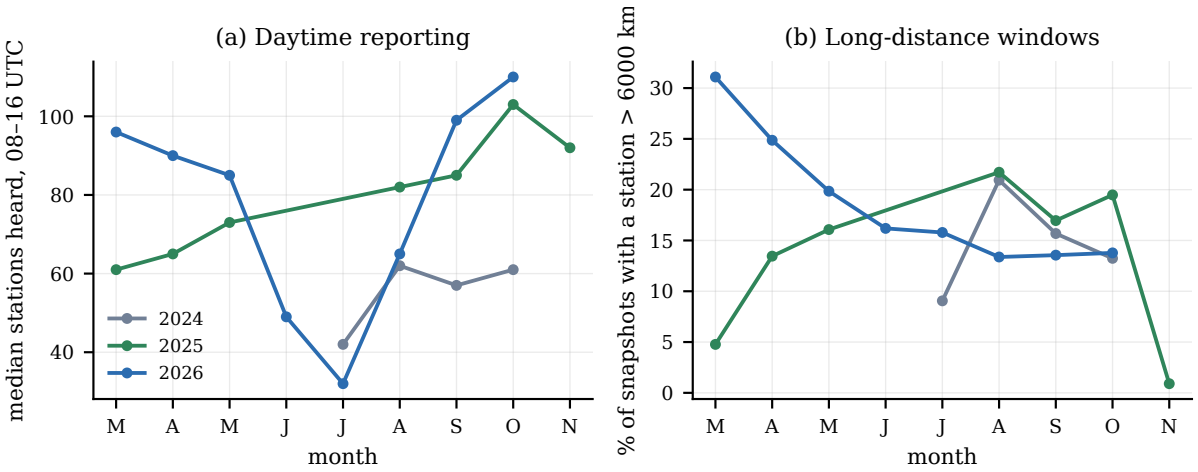


Figure 5. The seasons month by month: (a) median number of stations heard between 08 and 16 UTC; (b) share of snapshots with a station beyond 6 000 km. Only months with at least ten days of data are drawn.

Table 6. Monthly summary (months with at least five days of data). Daytime median = median number of stations heard between 08 and 16 UTC; stations = distinct callsigns heard in the month; DX share = snapshots with a station beyond 6 000 km. MUF and SFI exist only for the live-logged part of 2026 (from 2026-05-08).

Month	Days	Snapshots	Stations	Daytime median	DX share (%)	Median MUF (MHz)	Mean SFI
2024-07	24	2 363	468	42	9	–	–
2024-08	30	3 410	660	62	21	–	–
2024-09	13	1 454	457	57	16	–	–
2024-10	15	1 503	445	61	13	–	–
2025-03	26	2 642	546	61	5	–	–
2025-04	30	3 560	707	65	13	–	–
2025-05	28	3 565	704	73	16	–	–
2025-08	27	3 325	755	82	22	–	–
2025-09	30	3 603	798	85	17	–	–
2025-10	31	3 136	773	103	19	–	–
2025-11	14	1 331	476	92	1	–	–
2026-03	24	2 052	661	96	31	–	–
2026-04	28	2 920	723	90	25	–	–
2026-05	31	4 413	745	85	20	19.8	121
2026-06	28	3 762	599	49	16	18.8	139
2026-07	26	3 370	514	32	16	18.1	133
2026-08	26	2 737	642	65	13	19.9	117
2026-09	28	2 597	707	99	14	22.5	106
2026-10	28	2 526	765	110	14	24.4	110
2026-11	6	337	335	103	5	26.8	101

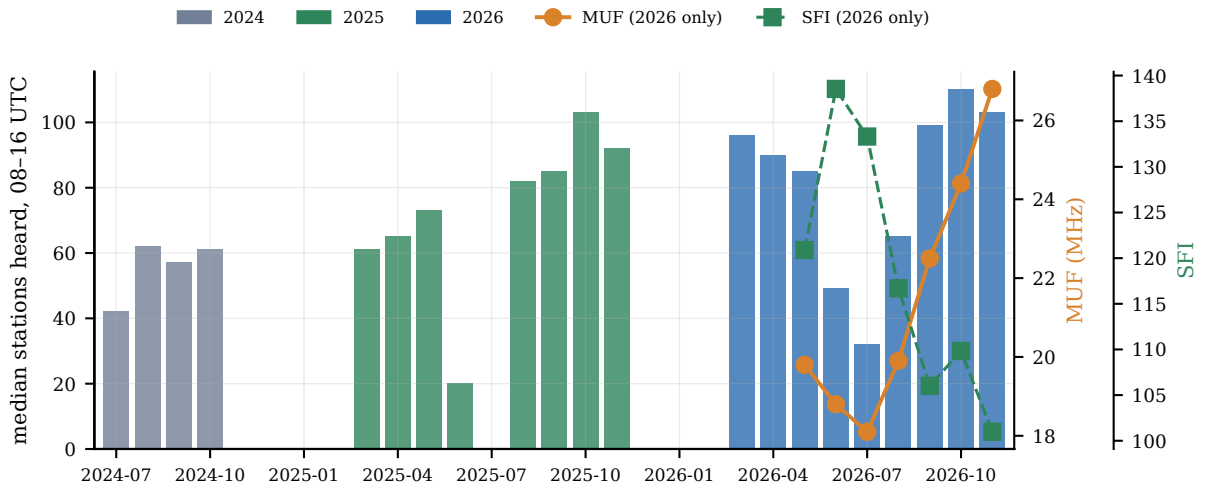


Figure 6. Monthly median number of stations heard between 08 and 16 UTC (bars, coloured by season) and, for the live-logged months of 2026 only, the median Juliusruh MUF over the same hours (circles) and the mean SFI (squares). November 2026 comprises six days.

4.7 Space weather

Figure 7 and Tables 7–9 relate the hour-adjusted reporting index and the DX share to the three indices recorded at each poll. Only the 166 live-logged days of 2026 (2026-05-08 to 2026-11-06, 18 885 snapshots) carry these readings, so this section describes one season.

MUF. The Juliusruh MUF is the strongest single predictor. At MUF below 12 MHz the index is only 27 and the DX share 4 %; at 20–24 MHz the index is 144 and the DX share 19 %. The raw correlation between reporters and MUF is 0.70 (95 % CI 0.67–0.74), but much of that is the common diurnal rhythm; after removing it the correlation with relative reporting is 0.43 (0.37–0.48). The relationship is therefore real but moderate: a high MUF is necessary rather than sufficient.

Geomagnetic activity. Reporting falls with the K-index, but gently: the index is 113 at $K = 0$ and 88 at $K \geq 5$; the correlation is -0.16 (-0.22 to -0.10). The DX share is far more sensitive: 22 % at $K = 0$, 19 % at $K = 1$ and only 7 % on average for $K \geq 3$. The A-index gives the same picture over the day: for $A < 8$ the DX share is 20 % (index 111), for $A \geq 30$ it is 2.7 % and the index 66. Disturbed conditions thus mostly remove the weakest, longest paths first, which is the expected behaviour of an auroral-zone path at 60°N.

Solar flux. The solar flux has no detectable effect on hourly reporting on its own ($r = 0.05$, CI -0.05 to 0.15; the raw value is -0.13). This is not a statement that the Sun is irrelevant: SFI changed only between about 89 and 203 during these days, strong flux comes with flares and storms (Section 4.8), and SFI is a daily number while reporting responds to the hour-to-hour ionosphere.

Table 7. Reporting versus Juliusruh MUF. Index: stations heard relative to the typical value for the same UTC hour (100 = typical); DX share: snapshots with a station beyond 6 000 km. Confidence intervals by day-block bootstrap.

MUF (MHz)	Snapshots	Mean stations	Index	95 % CI	DX share (%)	95 % CI
< 12	1 542	6	27	21–34	4	2–7
12–16	3 493	13	38	34–42	10	7–13
16–20	7 534	52	81	75–88	15	13–18
20–24	4 710	103	144	135–153	19	16–23
24–28	1 169	110	142	130–154	20	14–27
≥ 28	352	100	127	115–138	16	7–27

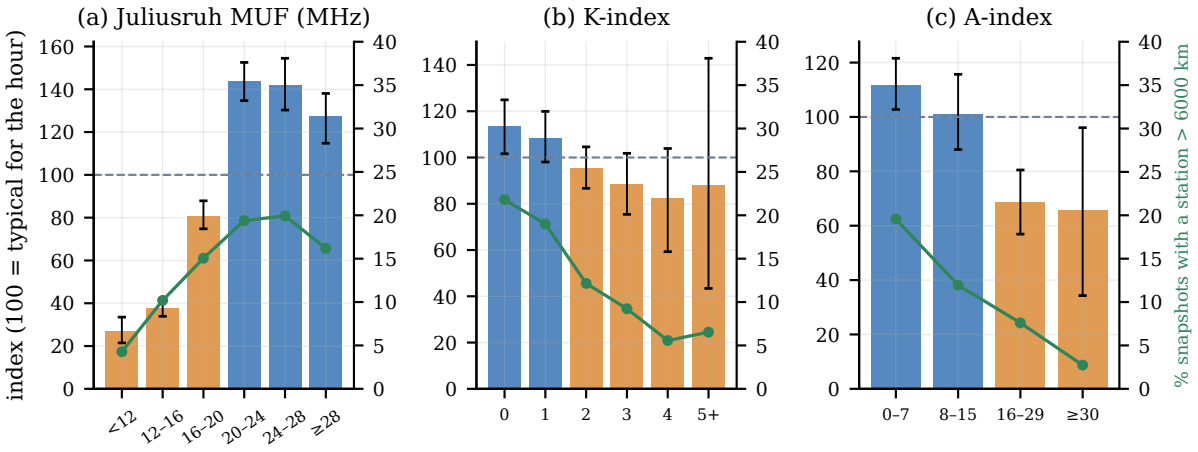


Figure 7. Hour-adjusted reporting index (bars, left axis, 100 = typical for the UTC hour; 95 % day-block bootstrap intervals) and DX share (green line, right axis) by (a) Juliusruh MUF, (b) K-index and (c) A-index. Blue bars are at or above the typical level, orange below.

Table 8. Reporting versus the K-index (day-block bootstrap intervals).

K-index	Snapshots	Mean stations	Index	95 % CI	DX share (%)	95 % CI
0	2 325	64	113	102–125	22	16–28
1	6 238	65	108	98–120	19	16–22
2	5 850	55	95	87–105	12	10–14
3	3 174	50	88	75–102	9	6–13
4	880	45	82	59–104	6	0–12
5+	398	51	88	43–143	7	1–15

Table 9. Reporting versus the A-index (day-block bootstrap intervals).

A-index	Snapshots	Mean stations	Index	95 % CI	DX share (%)	95 % CI
0–7	9 204	65	111	103–122	20	17–22
8–15	6 199	59	101	88–116	12	9–15
16–29	2 750	39	69	57–80	8	5–10
≥ 30	732	41	66	34–96	3	0–7

4.8 Case study: 24 June – 9 July

The most disturbed fortnight of the live-logged data is instructive (Table 10, Figure 8). Solar flux rose from 131 on 24 June to above 195 on 30 June–3 July with M-class flares on several days. Reporting followed no simple pattern: 2 and 3 July had the best days of the fortnight (about 93 and 70 stations, with K = 2–3) while 25 June (K = 5, A = 26) and 1 July (K = 4, A = 18) were weak. The worst day of the live-logged data was 4 July: K = 7, A = 74, and only 10 stations in the daytime window against the typical 80 (the lowest daytime mean of the whole archive, 8 stations, is from 2025-03-25, a day without solar data). The disturbance persisted: 5 July (K = 5, A = 74) had 13 stations, and 6 July, with K down to 3 and A = 18, still only 10; reporting recovered to 28 on 7 July and to 67 on 9 July. The pattern, a strong single-day collapse followed by a recovery lag of two to three days, is the classic ionospheric storm signature at high geomagnetic latitudes: it is the K-index and the A-index, not the X-ray class or the flux, that mark the days that failed.

Table 10. The period 24 June – 9 July 2026: space weather and reporting (daytime mean).

Date	SFI	Max K	Max A	Max X-ray	Stations 08–16 UTC
2026-06-24	131	3	15	C	37
2026-06-25	139	5	26	C	16
2026-06-26	157	4	26	C	24
2026-06-27	166	3	12	C	15
2026-06-28	188	2	8	C	39
2026-06-29	187	1	8	C	27
2026-06-30	196	4	18	M	40
2026-07-01	203	4	18	M	15
2026-07-02	201	2	17	M	93
2026-07-03	201	3	12	M	70
2026-07-04	184	7	74	M	10
2026-07-05	159	5	74	M	13
2026-07-06	138	3	18	C	10
2026-07-07	123	2	7	C	28
2026-07-09	115	4	16	C	67

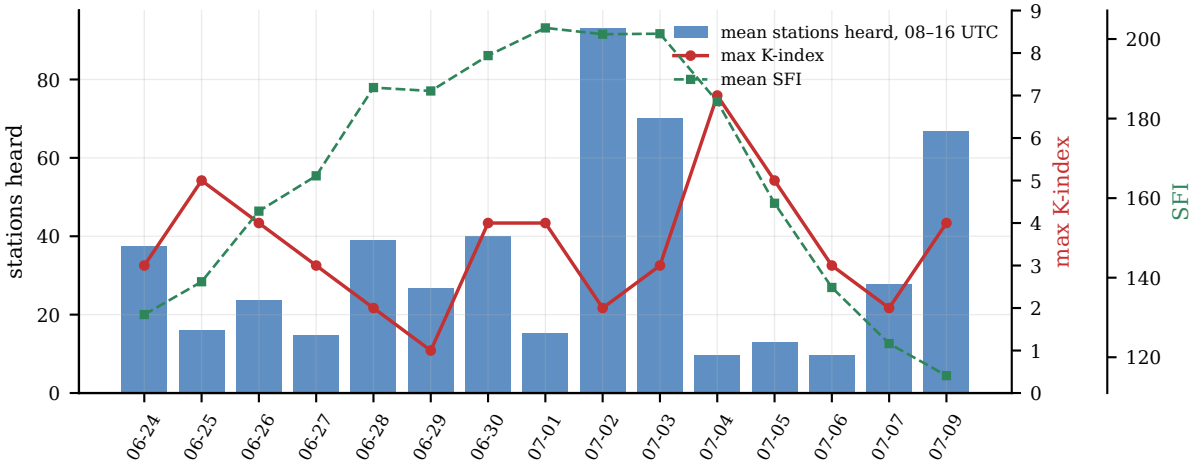


Figure 8. Daytime reporting (bars) with maximum K-index (red) and mean SFI (green) for 24 June – 9 July 2026 (8 July had no complete daytime data).

4.9 Records

All eight farthest stations (Table 11) are in Australia; the record is VK7JJ at 15 619 km on 2024-07-10, which corresponds to 78 095 km per watt of transmitter output. Three of the eight rows (`VK7JJ`, `VK7FLYN`, `VK7FLYN/2`) share one grid square and are probably one receiving site using several callsigns. Of the 28 stations that were heard beyond 12 000 km, 25 are in Oceania, two in South America (Argentina and Chile) and one in Antarctica (`DP0GVN/1`).

Table 11. The eight farthest stations (one row per station, quality-controlled).

#	Station	Region	Grid	Distance (km)	When (UTC)
1	VK7JJ	Australia	QE38lr	15 619	2024-07-10 17:38
2	VK7FLYN	Australia	QE38lr	15 619	2026-09-24 06:08
3	VK7FLYN/2	Australia	QE38lr	15 619	2026-09-30 05:48
4	VK7EKA	Australia	QE38kv	15 596	2024-10-22 06:28
5	VK3AG	Australia	QF31	15 385	2026-03-08 13:48
6	SWLVK3	Australia	QF22xv	15 208	2025-10-10 06:08
7	SWL/VK3	Australia	QF12wb	15 165	2026-06-14 21:38
8	VK2EFM	Australia	QF56oq	15 156	2024-08-16 19:48

5 Discussion

What the data support. The diurnal structure, the European concentration and the existence and timing of the North American, African and Oceanian windows are all robust: they rest on hundreds of days or thousands of snapshots and are not sensitive to the quality-control step. The dependence of reporting on MUF and on K/A is also robust in sign and, with day-block intervals, in size, but it rests on one season (2026). The seasonal pattern within a year is real in the data but cannot be attributed, and the differences between the seasons are mixed with changes in the receiver network and with the rebuild of the older snapshots.

Limits.

- *One path, one band, three summer seasons.* The archive covers July–October 2024, March–November 2025 and March–November 2026 on 14 MHz only. The car-mounted antenna is not used in winter (snow, low temperature), so no winter data exist and seasonal statements concern March–November. Whether the equipment, the antenna position or the operating habits were identical in the three seasons is not recorded in the data, which limits the comparison between seasons.
- *Rebuilt snapshots and missing space weather.* The snapshots of 2024, 2025 and early 2026 were reconstructed from `wspr.live` after the fact. The live poll could not see reports that arrived later, so the rebuilt snapshots could list more stations; a comparison of 995 live snapshots with their rebuild (Section 2.2) found a ratio of 1.001, i.e. no measurable difference, but only for one week in June 2026. No solar or MUF values were rebuilt, so the space-weather results are limited to 166 days.
- *Receiver network.* Reporting counts depend on how many receivers are listening. The 88 most frequent stations account for half of all reports, so a station going off-air shifts the counts without any change in propagation. The sparsely covered directions (north, east) say nothing about propagation there.
- *Mobile transmitter.* The car moves; 86 % of the logged transmissions come from the home square. Position-dependent effects (antenna surroundings, local noise at the receiving end of a path) are not separated. The companion paper on position reporting examines the movement itself.
- *Overlapping windows.* Consecutive snapshots overlap by 50 minutes; all intervals use day-block resampling for this reason, but the effective sample size is closer to the number of days than to the number of snapshots.
- *Indices versus path.* Juliusruh is about 900 km south-west of the transmitter and not on the path to North America; K and A are planetary values, and SFI is daily. The indices are proxies.

- *SNR.* Reported SNR is estimated by the receiving station and depends on its noise floor and software; only medians over many stations are interpreted.

6 Data quality and software notes

Two issues were found while preparing the study and are relevant to anyone reading the archive.

Under-reported Best DX. The logger's per-transmission record (used for the "Best DX" figure on the live page) was written from a single poll six minutes after the transmission. Distant stations upload to WSPR Net later, so the farthest reporter of a transmission was often missing; comparing the two databases day by day over the live-logged days showed that on 68 of 164 days (41 %) the farthest reporter in the replay data was farther than the stored value (largest difference on 2026-10-07: 4 553 km stored against 15 614 km). The poll now covers the last 45 minutes and raises the stored value as later reports arrive; the backfill script raised the stored values of the earlier seasons in the same way. This study uses the replay database.

Spurious reporters. Three callsign–grid pairs carried a distance that contradicted their own grid (Section 2.5); this is the kind of entry that the cleaning step is meant to catch, and the live page now applies the same check.

7 Conclusions

- The band is open (median at least 50 stations) from 7 to 20 UTC, with a median of about 75 stations at midday; the long-distance windows lie at the edges of the day, not in its middle.
- North America is reached most often in the morning (04–07 UTC), Africa and Oceania in the afternoon and evening (16–19 UTC); the farthest reception was 15 619 km.
- 73 % of days had a reporter beyond 6 000 km with 200 mW and a car-roof whip.
- Within the day, the MUF matters more than anything else the logger records; K and A matter most for the longest paths; the daily solar flux has no detectable effect.
- Midsummer is the weakest period (July 2024 and July 2026) and autumn the strongest (October 2025 and 2026), independently of the solar flux; for the same months, daytime reporting was higher in 2025 and 2026 than in 2024.

A Reproducing the analysis

The figures, tables and every number in the text are produced from the two SQLite databases by the script `doc/src/analyze.py` (Python 3 with NumPy and Matplotlib; the standard library for everything else). It writes `numbers.tex`, one `tab_*.tex` file per table and the PDF figures to `doc/src/out`; the paper is then built with `pdflatex propagation_study.tex` (twice). The databases themselves are not part of the repository.

Version history

Version	Date	Changes
1.0	6 November 2026	First version: 2026 season only, 166 days.
1.1	6 November 2026	Two earlier seasons (2024, 2025) rebuilt from `wspr.live`: 2024-07-04 to 2026-11-06, 54 868 snapshots; new section on the seasons; space-weather results explicitly limited to the live-logged days of 2026.
1.2	6 November 2026	Measured the effect of late reports on the rebuilt seasons: live and rebuilt snapshots agree (ratio 1.001).

References

[1] WSPRnet — the Weak Signal Propagation Reporter Network. `https://wsprnet.org`

[2] wspr.live — ClickHouse interface to the WSPRnet spot database. `https://wspr.live`

[3] J. Taylor, K1JT, *WSJT-X User Guide* (WSPR mode) and *WSPR 2.0 User's Guide*. `https://wsjt.sourceforge.io`

[4] ZachTek, *WSPR Desktop Transmitter* documentation. `https://www.zachtek.com/1012`

[5] Global Ionospheric Radio Observatory, Digital Ionogram Database (DIDBase). `https://giro.uml.edu/didbase/`

[6] Leibniz Institute of Atmospheric Physics, Juliusruh Space Weather Monitor. `https://ionosonde.iap-kborn.de/actuellz.htm`

[7] NOAA Space Weather Prediction Center, solar and geophysical indices. `https://www.swpc.noaa.gov`

[8] N0NBH, solar-terrestrial data feed. `https://www.hamqsl.com/solar.html`